

An Explicit 26-Measurement Phase-Retrieving Frame in $\mathbb{C}^8$

A five-parameter family and its exact boundary
Proof Candidate – Version 2

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Abstract

We construct a five-parameter family of 26-measurement frames in $\mathbb{C}^8$. For real parameters b, c, s, m, γ with $0 < s < 1$, $m < 0$, and $\gamma > 0$, the family gives phase retrieval precisely when $c > b^2/s$. The measurements determine squared moduli of polynomial evaluations at any 26 distinct positive real points. Three vanishing coefficients and a bound on positive roots imply that these values determine the squared-modulus polynomial. To prove uniqueness up to phase, we compare the average and difference of two possible signals. Coefficient elimination reduces the problem to positive quadratic expressions and an invertible two-by-two matrix. When $c \leq b^2/s$, we construct nonproportional polynomials with identical squared moduli. The family contains the frame from the first release and, after an invertible change of signal coordinates, yields a representative whose defining polynomials contain twenty monomial terms in total. The result gives $m_{\mathbb{C}}(8) \leq 26$; minimality is not established.

Status. Proof candidate; not peer reviewed. Section 8 describes the completed checks and their scope.

1 The question, and a short explicit answer

A measurement of $x \in \mathbb{C}^8$ is a real number $|q^*x|^2$, where $q^*x = \sum_{k=0}^7 \overline{q_k}x_k$. Multiplying x by a complex scalar of modulus one leaves every such measurement unchanged. We ask whether 26 choices of q distinguish vectors in $\mathbb{C}^8$ up to this unavoidable common phase. A family with this property is called a phase-retrieving frame. For any dimension d , let $m_{\mathbb{C}}(d)$ denote the least number of such measurements needed to distinguish all vectors in $\mathbb{C}^d$ up to a common phase.

Consider the following eight polynomials:

$$\begin{aligned} r_0 &= 1 + it + it^5 - it^{13}, & r_1 &= t^2 + \frac{i}{2}t^3 - it^{11}, \\ r_2 &= t^4 + it^5 - it^9, & r_3 &= t^6 - it^7, \\ r_4 &= t^8 + it^9 + it^{13}, & r_5 &= t^{10} + \frac{i}{2}t^{11}, \\ r_6 &= t^{12} + it^{13}, & r_7 &= t^{14}. \end{aligned} \tag{1}$$

For $j = 1, \dots, 26$ take

$$q_j = 2(\overline{r_0(j)}, \dots, \overline{r_7(j)})^T. \tag{2}$$

Every entry is a Gaussian integer. The proof below shows that equality of these 26 measurements forces two vectors to differ by a common phase. The eight defining polynomials in (1) contain twenty monomial terms in total.

The geometric idea is to represent a signal by a polynomial f . With our normalization, the measurement at t_j is $4|f(t_j)|^2$ and therefore determines $|f(t_j)|^2$. We arrange two properties: 26 samples determine this whole real polynomial, and the polynomial determines f up to phase. For the second property, align their lowest-degree nonzero coefficients at $t = 0$ and write them as $v + h$ and $v - h$. Equal squared moduli mean that $v(t)$ and $h(t)$ are perpendicular in the real plane $\mathbb{C}$ at every real t . The chosen space makes this possible only when $h = 0$.

We work with a larger family to identify the sign condition that separates uniqueness from ambiguity. The basis in (1) is obtained from the parameter choice $(b, c, s, m, \gamma) = (0, 1, \frac{1}{2}, -1, 1)$ by an invertible linear transformation, given in Section 7. The frame in the first release is another member of the family.

Conca, Edidin, Hering, and Vinzant proved that $4d - 4$ generic complex measurements suffice in dimension d [1]. Vinzant constructed eleven measurements in dimension four [2]. Here we give explicit 26-measurement frames in dimension eight and characterize the successful parameters within our family.

2 A family with a sharp boundary

For a polynomial $f(t) = \sum c_j t^j$, put $f^\#(t) = \sum \bar{c}_j t^j$. Thus $ff^\#$ has real coefficients and, for real t , equals $|f(t)|^2$. We write $[t^j]f$ for the coefficient of t^j in f ; for a matrix of polynomials this notation is applied entry by entry. Conjugation also acts entry by entry, and $a^{\#T}$ means $(a^\#)^T$. Let $b, c \in \mathbb{R}$ and assume

$$0 < s < 1, \quad m < 0, \quad \gamma > 0. \quad (3)$$

Define

$$a = \begin{pmatrix} 1 + it + bt^3 + ict^5 \\ t^2 + ist^3 - bt^5 \\ t^4 + it^5 \end{pmatrix}, \quad J = \begin{pmatrix} 2c & 0 & 2 \\ 0 & 2s & 0 \\ 2 & 0 & 0 \end{pmatrix}. \quad (4)$$

Since $\det J = -8s$, the matrix is invertible on the whole domain, including $c \leq b^2/s$. Concatenate the following entries to form $p = (p_0, \dots, p_7)^T$:

$$p = (a + \gamma t^8 J^{-1} a^\#, \quad t^6 + imt^7, \quad t^8 a, \quad t^{14})^T. \quad (5)$$

The transpose in this notation only indicates a column of eight entries. Let $U = \text{span}_{\mathbb{C}}\{p_0, \dots, p_7\}$. Each p_j begins with t^{2j} , with coefficient one, so these entries are linearly independent and $\dim_{\mathbb{C}} U = 8$.

The identity $[t^5](aa^T) = iJ$ explains the choice of J^{-1} : the conjugated correction cancels the degree-thirteen cross terms, as shown in Section 3. Without this correction, the space would contain both a_0 and $t^8 a_0$, and hence the ambiguous pair $a_0(1 + it^8)$, $a_0(1 - it^8)$.

Theorem 1. *Under (3), let $t_1, \dots, t_{26}$ be any distinct positive real numbers and set $q_j = 2\overline{p(t_j)}$. The vectors q_j give phase retrieval if and only if*

$$\Delta := c - \frac{b^2}{s} > 0. \quad (6)$$

When $\Delta \leq 0$, the space U contains two nonproportional polynomials with identical squared-modulus polynomials. Consequently, no set of real sampling points can distinguish all elements of U up to phase by their squared moduli.

The quantity Δ is the minimum of the quadratic expression R in (11). Positivity of R is the key sign condition in the uniqueness proof. Section 6 constructs ambiguous pairs when $\Delta \leq 0$.

For $f_x = \sum x_k p_k$ one has $q_j^* x = 2f_x(t_j)$. Thus the measurement is $|q_j^* x|^2 = x^*(q_j q_j^*)x$, with the rank-one matrix $q_j q_j^*$. Every q_j is nonzero since $p_7(t_j) = t_j^{14} > 0$. When the parameters and sampling points are rational, a common integer scaling makes every entry of every measurement vector a Gaussian integer.

3 Why 26 positive sampling points suffice

All polynomials in U have degree at most 14. For every $u \in U$,

$$[t]uu^\# = [t^5]uu^\# = [t^{13}]uu^\# = 0. \quad (7)$$

The seed satisfies $[t](aa^{\#T}) = [t^5](aa^{\#T}) = 0$ and $[t^5](aa^T) = iJ$; these follow by multiplying the three displayed entries in (4). The correction starts in degree eight and hence does not affect the first two missing degrees. In products $p_j p_k^\#$ with $j, k < 3$, the correction contributes the following matrix to the coefficient of degree thirteen:

$$\gamma(iJJ^{-T} - iJ^{-1}J) = 0.$$

Products between a corrected seed entry and an entry of $t^8 a$, with coefficient conjugation on one factor, reduce to the vanishing degree-five coefficient of $aa^{\#T}$. The squared modulus of $p_3 = t^6 + imt^7$ is $t^{12} + m^2 t^{14}$. Products involving p_3 and an entry of the corrected seed or $t^8 a$ have no term of degree thirteen; all other products start in higher degree. This proves (7) for all products $p_j p_k^\#$ and hence for every u .

Lemma 2. *A nonzero real polynomial with at most M nonzero monomials has at most $M - 1$ distinct positive roots.*

Proof. Induct on the actual number $N \leq M$ of nonzero monomials. The case $N = 1$ is immediate. For $N \geq 2$, divide by the lowest power of t , which preserves positive roots. The resulting polynomial has a nonzero constant term and $N - 1$ other monomials. Its derivative is nonzero and has $N - 1$ monomials, so by induction it has at most $N - 2$ distinct positive roots. Rolle's theorem gives at most $N - 1 \leq M - 1$ distinct positive roots for the original polynomial. $\square$

For $f, g \in U$, the real polynomial $ff^\# - gg^\#$ has degree at most 28. Of the 29 possible coefficients, three vanish by (7). Thus it has at most 26 nonzero monomial terms. If 26 sampled squared moduli agree, Lemma 2 gives

$$ff^\# = gg^\#. \quad (8)$$

The use of distinct *positive* nodes is essential to this argument. It remains to understand when (8) forces a common phase.

4 Comparing an average with its difference

If one polynomial in (8) is zero, so is the other. For $f \neq 0$, let i be the least index with a nonzero coefficient in its expansion in the basis $p_0, \dots, p_7$. If the first term of f at zero is λt^{2i} , the first term of $ff^\#$ is $|\lambda|^2 t^{4i}$. Equation (8) therefore gives the same initial index i for g , with

initial coefficient of modulus $|\lambda|$. Rotate the phases of f and g independently and divide both by this common modulus so that both initial coefficients become one. Then

$$\begin{aligned} v &= (f + g)/2 = p_i + \sum_{j>i} (x_j + iy_j)p_j, \\ h &= (f - g)/2 = \sum_{j>i} (r_j + i\alpha_j)p_j, \\ F &= hv^\# + h^\#v = 0. \end{aligned} \tag{9}$$

Since $ff^\# - gg^\# = 2(hv^\# + h^\#v)$, equality of the squared-modulus polynomials is equivalent to $F = 0$. Here $x_j, y_j, r_j, \alpha_j \in \mathbb{R}$ are coordinates of the normalized average and difference. For real t , $F(t) = 2 \operatorname{Re}(h(t)\overline{v(t)})$, so $F = 0$ expresses pointwise orthogonality in the real plane $\mathbb{C}$. The normalization divides only by the nonzero initial coefficient modulus and imposes no restriction on measurement zeros, common factors, or repeated roots.

For each fixed v , the map $h \mapsto hv^\# + h^\#v$ is real-linear on $\operatorname{span}_{\mathbb{C}}\{p_{i+1}, \dots, p_7\}$, viewed as a real vector space. We prove its injectivity for every allowed v by triangular coefficient elimination, followed by scalar equations with strictly signed factors and a two-dimensional linear system in the last case.

The expansions at $t = 0$ are

$$p_j = t^{2j}(1 + i\sigma_j t) + O(t^{2j+2}), \quad (\sigma_0, \dots, \sigma_7) = (1, s, 1, m, 1, s, 1, 0).$$

If $h \neq 0$ and $j > i$ is its first nonzero index, the first two possible coefficients of F are $2r_j$ and, after $r_j = 0$, $2(\sigma_i - \sigma_j)\alpha_j$. Higher indices cannot contribute to them. It follows that $r_j = 0$ and $\sigma_i = \sigma_j$. The four values $1, s, m, 0$ are distinct under (3), so the only possible pairs are

$$(i; j) = (0; 2, 4, 6), (1; 5), (2; 4, 6), (4; 6). \tag{10}$$

For $i = 3, 5, 6, 7$ there is no possible nonzero difference. Thus the argument also covers signals whose first several p -basis coordinates vanish.

Write $E_k = [t^k]F$. Eliminate the difference coefficients in increasing order of j . The coefficient of F at degree $2i + 2j$ determines r_j by division by 2. After that substitution, the next coefficient determines α_j by division by the fixed nonzero number $2(\sigma_i - \sigma_j)$ whenever the slopes differ. Equal slopes leave α_j free and the corresponding equation as a constraint. The term involving α_j in its even-degree equation is zero because the leading coefficients are real. No later index can affect these equations. The recurrence uses only these fixed nonzero divisors. Appendix A records all substitutions needed below.

5 The positive side of the boundary

Assume now $\Delta > 0$. Put

$$\begin{aligned} R(X, Y) &= c - 2bY + s(1 - s)X^2 + sY^2 \\ &= \Delta + s(1 - s)X^2 + s(Y - b/s)^2 > 0, \\ D &= c + \gamma - \frac{b^2}{s - m} = \Delta + \gamma + \frac{(-m)b^2}{s(s - m)} > 0. \end{aligned} \tag{11}$$

These inequalities force the remaining difference coefficients in (10) to vanish. Every equation below is a coefficient of the same F after the stated preceding substitutions.

5.1 The average starts with p_4 or p_1

For $i = 4$, the only possible first difference is at 6. Eliminating through degree 23 gives

$$E_{25} = 2\alpha_6 R(x_5, y_5).$$

Thus $\alpha_6 = 0$ and all remaining coefficients of h vanish. For $i = 1$, start the difference at 5 and eliminate through degree 17. Then

$$E_{21} = -2\alpha_5 \left[\gamma(1 + x_2^2 + y_2^2) + \frac{m(m-s)}{s} x_3^2 - m y_3^2 \right].$$

The bracket is at least $\gamma > 0$, so again $h = 0$.

5.2 The average starts with p_2

Start the difference at 4 and eliminate through degree 16. One obtains

$$E_{17} = -2\alpha_4 \left[\frac{(1-m)(s-m)}{1-s} x_3^2 + (s-m) \left(y_3 - \frac{b}{s-m} \right)^2 + D \right].$$

The coefficients of the squared terms are positive, and $D > 0$, so $\alpha_4 = 0$. The substitutions make all coefficients of h below index 6, and also r_6 , zero. Any remaining difference starts with $i\alpha_6 p_6$. Restarting at 6 and eliminating through degree 19 yields

$$E_{21} = -2\alpha_6 \left[\gamma + m(m-1)x_3^2 - m y_3^2 \right].$$

The bracket is at least γ , which forces $\alpha_6 = 0$ and $h = 0$.

5.3 The average starts with p_0 : removing the first difference

Start the difference at 2 and eliminate through degree 8. Then

$$E_9 = 2\alpha_2 \left[\frac{(1-s)(s-m)}{1-m} x_1^2 + (s-m) \left(y_1 - \frac{b}{s-m} \right)^2 + D \right].$$

This forces $\alpha_2 = 0$, and every coefficient of h before index 4 vanishes. The remaining imaginary coefficients are α_4 and α_6 .

5.4 The final two-dimensional system

For $i = 0$ and h starting at 4, degrees 8 through 15 give

$$\begin{aligned} r_4 &= 0, & r_5 &= -\alpha_4 y_1, \quad \alpha_5 = \alpha_4 x_1, \\ r_6 &= -\alpha_4 y_2, \\ r_7 &= \alpha_4 x_2 y_1 - \alpha_4 y_3 - \alpha_6 y_1, \\ r_7 &= (1-m)\alpha_4 x_3 + (s-1)\alpha_4 x_1 x_2 + (1-s)\alpha_6 x_1. \end{aligned} \tag{12}$$

Set

$$\begin{aligned} \rho &= \alpha_6 - \alpha_4 x_2, & R &= R(x_1, y_1), \\ T &= m((1-s)x_1 x_3 + y_1 y_3) - \gamma x_2, \\ S &= -\gamma(1 + x_1^2 + y_1^2 + x_2^2 + y_2^2) - m(m-1)x_3^2 + m y_3^2. \end{aligned} \tag{13}$$

Direct multiplication after (12) gives

$$\begin{aligned} E_{17} &= 2(R\rho + T\alpha_4), & E_{21} &= 2(S\alpha_4 + T\rho), \\ \begin{pmatrix} S & T \\ T & R \end{pmatrix} \begin{pmatrix} \alpha_4 \\ \rho \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} E_{21} \\ E_{17} \end{pmatrix}. \end{aligned} \quad (14)$$

The quadratic form is positive on the second coordinate axis and negative on the first: $R \geq \Delta > 0$, $S \leq -\gamma < 0$. In two dimensions the off-diagonal term contributes $-T^2$ to the determinant, so

$$RS - T^2 \leq -\gamma\Delta < 0. \quad (15)$$

The matrix is therefore invertible for every average v . Equivalently, with no division by its determinant,

$$RE_{21} - TE_{17} = 2\alpha_4(RS - T^2).$$

Since $E_{17} = E_{21} = 0$, $\alpha_4 = 0$ and then $\rho = \alpha_6 = 0$. Equation (12) makes every other difference coefficient zero.

All eight initial indices are now covered. Every solution of $F = 0$ satisfies the coefficient equations used above, and these equations force $h = 0$. Undoing the normalization shows that the original polynomials differ by a common unit phase. Together with Section 3, this proves the sufficient direction of Theorem 1.

6 Ambiguous pairs on and beyond the boundary

The last four entries of (5) span $t^8 U_4$, where

$$U_4 = \text{span}_{\mathbb{C}}\{a_0, a_1, a_2, t^6\}.$$

For $c \leq b^2/s$, we seek $v, h \in U_4$ such that $hv^\# + h^\#v = 0$ and $v + h$ is not proportional to $v - h$. Multiplying this pair by t^8 gives an ambiguous pair in U , independently of m and γ .

Start at the boundary $c = b^2/s$. The polynomials $v = a_0 + i(b/s)a_1$ and $h = ia_2 - (b/s)t^6$ belong to U_4 and simplify to

$$v = 1 + it + i\frac{b}{s}t^2, \quad h = it^4v, \quad f_{\pm} = v(1 \pm it^4).$$

On the real line, the factors $1 + it^4$ and $1 - it^4$ have identical moduli. This gives an explicit ambiguity at the boundary.

To reach $c < b^2/s$, we adjust the average and difference while keeping both in U_4 . The coefficients W, K, Z below are chosen so that every coefficient of $hv^\# + h^\#v$, except those of degrees nine and twelve, vanishes identically. We solve the remaining two equations on a real curve in the (X, Y) -plane. For real X, Y put

$$\begin{aligned} W &= b + sXY - sY, & K &= -bX + (s-1)WX - W, \\ Z &= bsXY + c(1-s)X + WY, \\ v &= a_0 + (X + iY)a_1 + iWa_2 + (Z + iK)t^6, \\ h &= ia_2 + (-Y + i(1-s)X)t^6. \end{aligned}$$

Define

$$\begin{aligned} R_0 &= c - 2bY + s(1-s)X^2 + sY^2, \\ G &= (b(2-s) + s^2Y)X + b - sY, & H &= (s-1)X - Y^2. \end{aligned}$$

Direct multiplication gives the full polynomial identity

$$hv^\# + h^\#v = 2R_0t^9 + 2(HG - (1-s)XYR_0)t^{12}. \quad (16)$$

By (16), $R_0 = G = 0$ implies $hv^\# + h^\#v = 0$.

If $b = 0$, take $Y = 0$ and $X = \sqrt{-c/(s(1-s))}$. This is real for $c \leq 0$ and gives $R_0 = G = 0$. If $b \neq 0$, solving $G = 0$ for Y on $0 \leq X < 1/s$ gives the curve

$$0 \leq X < 1/s, \quad Y = \frac{b((2-s)X + 1)}{s(1-sX)}.$$

Along this curve, $G = 0$ and

$$R_0(X) = c - \frac{b^2}{s} + s(1-s)X^2 + \frac{4b^2X^2}{s(1-sX)^2}. \quad (17)$$

For $c = b^2/s$ choose $X = 0$. For $c < b^2/s$, this expression is negative at zero, tends to $+\infty$ as $X \rightarrow (1/s)^-$, and has derivative

$$2s(1-s)X + \frac{8b^2X}{s(1-sX)^3} > 0 \quad (0 < X < 1/s).$$

Thus, when $b \neq 0$ and $c < b^2/s$, there is a unique root in $0 < X < 1/s$. If b, c, s are rational, that root is specified by the quartic equation

$$(s^2(1-s)X^2 + sc - b^2)(1-sX)^2 + 4b^2X^2 = 0$$

together with the isolating interval $0 < X < 1/s$.

The resulting pair $v+h, v-h$ has the same squared-modulus polynomial by (16). Both have constant coefficient one, whereas h has first term it^4 . They cannot differ by a constant phase: the constant coefficients would force that phase to be one, contradicting $h \neq 0$. Multiplying by t^8 proves the necessary direction of Theorem 1.

7 Two explicit frames

7.1 Recovering the first released frame

At

$$(b, c, s, m, \gamma) = (1, 4, \frac{1}{2}, -2, 1)$$

formula (5) gives the eight polynomials of the first release. In particular $\Delta = 2 > 0$. The convention $q_j = 2\overline{p(j)}$ is unchanged. Thus the family theorem recovers the original frame as well as its phase-retrieval conclusion. The exact archived input is retained with the verification data in Section 8.

7.2 The twenty-term representative

At $(b, c, s, m, \gamma) = (0, 1, \frac{1}{2}, -1, 1)$, one has $\Delta = 1 > 0$. From its assembly basis p form

$$r_0 = p_0 - \frac{1}{2}p_6, \quad r_1 = p_1 - p_5, \quad r_2 = p_2 - \frac{1}{2}p_4 + \frac{1}{2}p_6, \quad r_j = p_j \quad (3 \leq j \leq 7).$$

This is a real triangular basis change with determinant one and gives exactly (1). To check its effect on signals, write $r = Ap$. Then $\sum x_k r_k = (A^T x)^T p$, and A^T is invertible. Uniqueness up to phase in the p coordinates therefore implies the same uniqueness in the r coordinates. The resulting frame is different from the first released one, even though both prove the same upper bound.

Corollary 3. *The explicit Gaussian-integer frame (2) gives $m_{\mathbb{C}}(8) \leq 26$.*

For either successful example, an equivalent matrix statement is useful. Let $\mathcal{A}_Q(H) = (q_j^* H q_j)_{j=1}^{26}$ on Hermitian matrices. The frame vectors $q_1, \dots, q_{26}$ span $\mathbb{C}^8$: a perpendicular vector would give a polynomial of degree at most 14 vanishing at 26 distinct points, hence the zero polynomial. If $H \geq 0$ lies in the kernel, each $q_j^* H q_j = 0$ implies $H q_j = 0$, so spanning gives $H = 0$. The same holds for $H \leq 0$. An indefinite rank-two Hermitian matrix is $xx^* - yy^*$ for nonproportional x, y by spectral decomposition, and the phase-retrieval property excludes it. Thus

$$\ker \mathcal{A}_Q \cap \{H = H^* : 0 < \text{rank } H \leq 2\} = \emptyset.$$

Conversely, if x, y have equal measurements but do not differ by a common phase, then $H = xx^* - yy^*$ is a nonzero Hermitian matrix of rank at most two in $\ker \mathcal{A}_Q$. Indeed, $xx^* = yy^*$ would force x, y to differ by a unit-modulus scalar, including the zero-vector case. Thus the kernel condition and phase retrieval are equivalent.

8 Scope and verification

Theorem 1 classifies the family (5) in the domain $0 < s < 1$, $m < 0$, $\gamma > 0$ and includes the original frame. The measurement bound remains $m_{\mathbb{C}}(8) \leq 26$. The minimum number of measurements, including whether 25 suffice, is not determined here. We make no assertion of generic phase retrieval with 26 measurements. The positivity estimates used in the proof are expressed after signal-dependent normalization and do not provide a global stability constant or a noise-recovery guarantee.

The proof uses polynomial algebra, Rolle’s theorem, and elementary linear algebra.

Three scripts using SymPy check the coefficient identities, the specializations, the basis transformation, and the frame entries. The family and boundary scripts provide the first two check groups below. The third script was transcribed separately and imports neither the original family checker nor its basis data. Together the checks cover the three missing degrees in all 64 ordered products $p_j p_k^\#$, all 28 pairs $0 \leq i < j \leq 7$, the elimination and boundary identities, and all 416 entries across the two frames. The execution logs record the following completed checks:

Check group	Checks completed
Original family checker	224
Boundary-identity checker	11
Separate transcription	847

The inequalities and the logical arguments covering all cases, restarts, and nonproportionality are given in the text. The exact frame data are in

- `data/v1_frame_gaussian_integer.json` for the original example;
- `data/v2_sparse_frame_gaussian_integer.json` for the twenty-term example.

The research, exposition, and computational checks were developed with substantial AI assistance. External expert review and proof-assistant verification have not been completed.

Version history. The first version of this manuscript was released on 19 September 2026 and is archived at [doi:10.5281/zenodo.22847200](https://doi.org/10.5281/zenodo.22847200). This revision extends that construction to the family studied here.

A The coefficient substitutions

The lists below record the substitutions used in Section 5. Each list assumes that the p -basis coefficients of h below the stated index vanish. The recurrence in Section 4 determines the displayed components; components not yet determined remain free at that stage. All denominators depend only on the fixed parameters and are nonzero under (3). The final case $(i, j) = (0, 4)$ appears in (12).

$i = 4$, first difference at 6: degrees 20 through 23

$$r_6 = 0, \quad r_7 = -\alpha_6 y_5, \quad \alpha_7 = (1 - s)\alpha_6 x_5.$$

$i = 1$, first difference at 5: degrees 12 through 17

$$\begin{aligned} r_5 &= 0, & r_6 &= -\alpha_5 y_2, & \alpha_6 &= \alpha_5 x_2, \\ r_7 &= -\alpha_5 y_3, & \alpha_7 &= \frac{s - m}{s} \alpha_5 x_3. \end{aligned}$$

$i = 2$, first difference at 4: degrees 12 through 16

$$\begin{aligned} r_4 &= 0, & r_5 &= -\alpha_4 y_3, & \alpha_5 &= \frac{1 - m}{1 - s} \alpha_4 x_3, \\ r_6 &= \alpha_4 \left[-b + (s - m)y_3 - \frac{s - m}{1 - s} x_3 y_3 - y_4 \right]. \end{aligned}$$

$i = 2$, restarted difference at 6: degrees 16 through 19

$$r_6 = 0, \quad r_7 = -\alpha_6 y_3, \quad \alpha_7 = (1 - m)\alpha_6 x_3.$$

$i = 0$, first difference at 2: degrees 4 through 8

$$\begin{aligned} r_2 &= 0, & r_3 &= -\alpha_2 y_1, & \alpha_3 &= \frac{1 - s}{1 - m} \alpha_2 x_1, \\ r_4 &= \alpha_2 \left[b + (m - s)y_1 + \frac{s - m}{1 - m} x_1 y_1 - y_2 \right]. \end{aligned}$$

In the cases $(i, j) = (2, 4)$ and $(0, 2)$, these formulas leave the higher coordinates free. If $\alpha_4 = 0$ in the first case, all difference coefficients below index 6 vanish; if $\alpha_2 = 0$ in the second, all coefficients below index 4 vanish. The argument can therefore restart at those indices, as in Section 5.

References

- [1] A. Conca, D. Edidin, M. Hering, and C. Vinzant, *An algebraic characterization of injectivity in phase retrieval*, Applied and Computational Harmonic Analysis 38 (2015), 346–356. [doi:10.1016/j.acha.2014.06.005](https://doi.org/10.1016/j.acha.2014.06.005).
- [2] C. Vinzant, *A small frame and a certificate of its injectivity*, 2015 International Conference on Sampling Theory and Applications, 197–200. [arXiv:1502.04656](https://arxiv.org/abs/1502.04656).